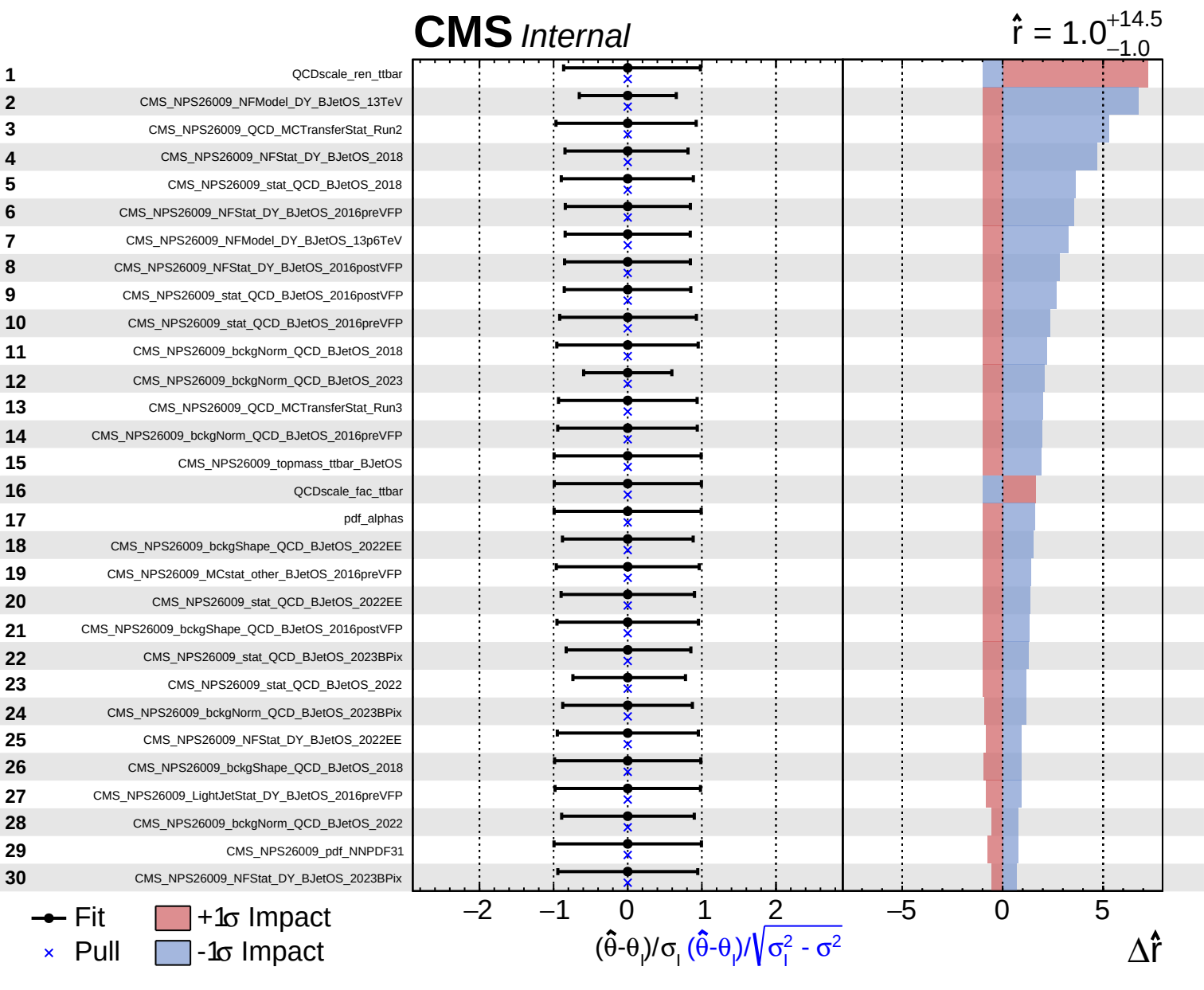
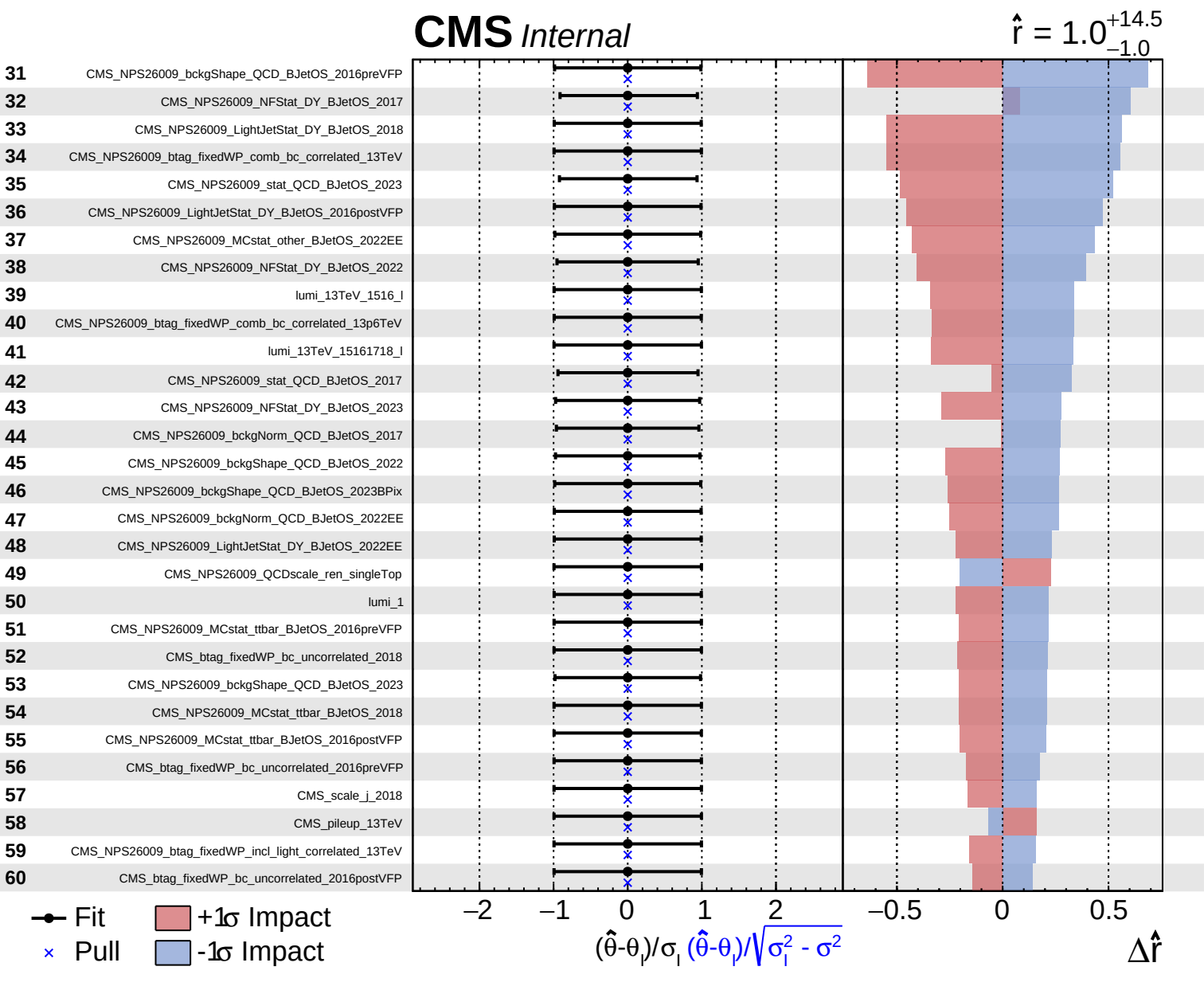
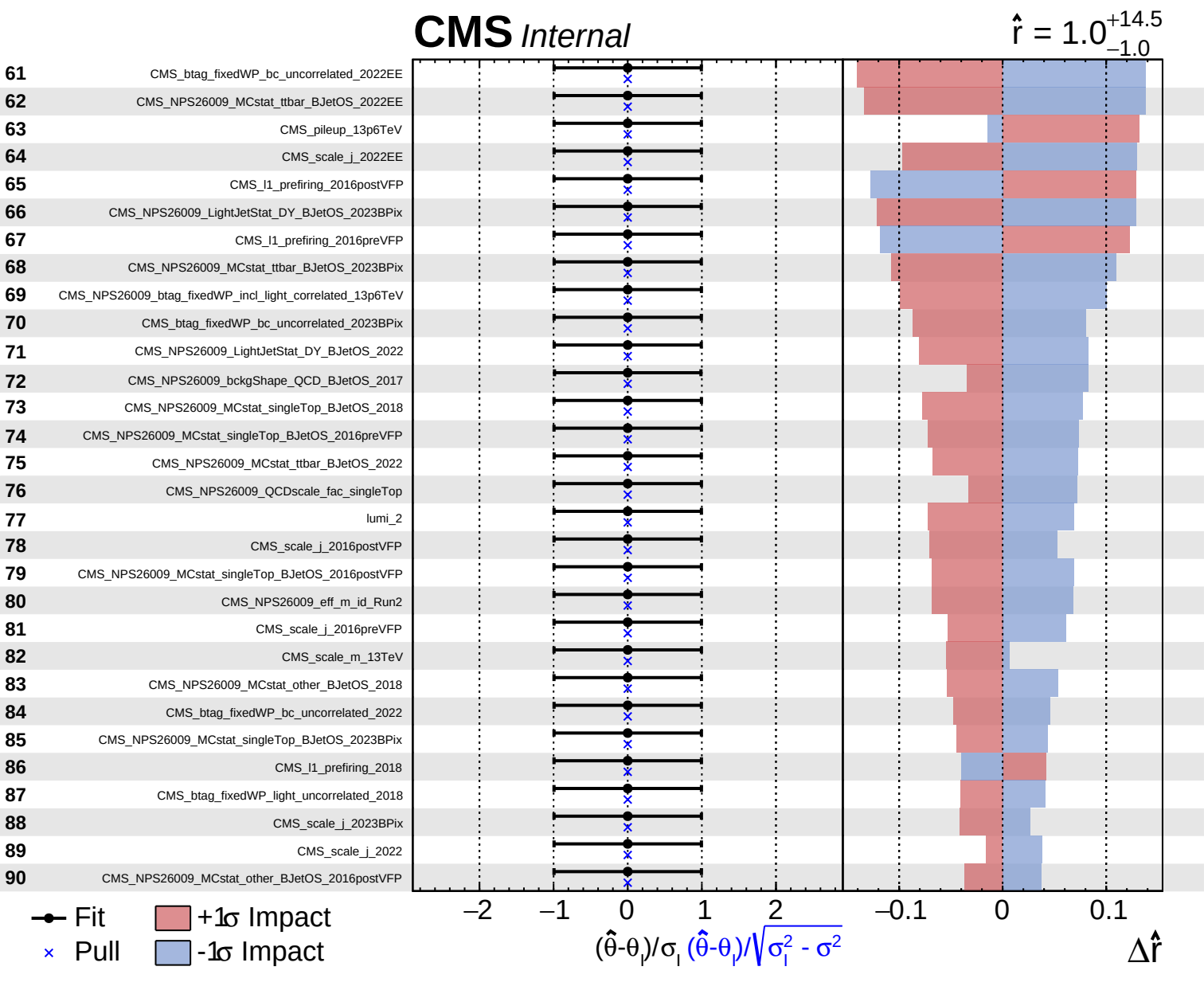
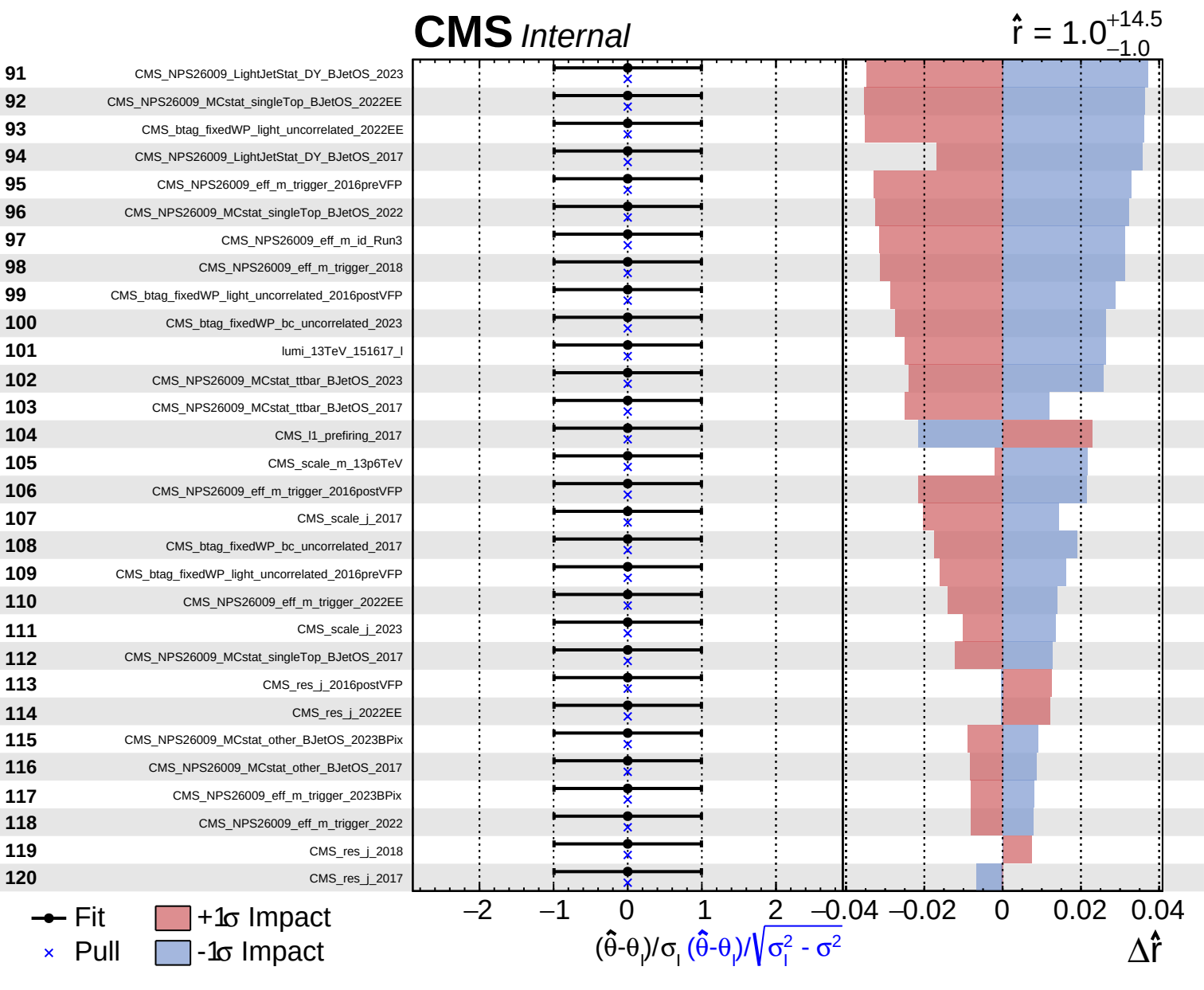
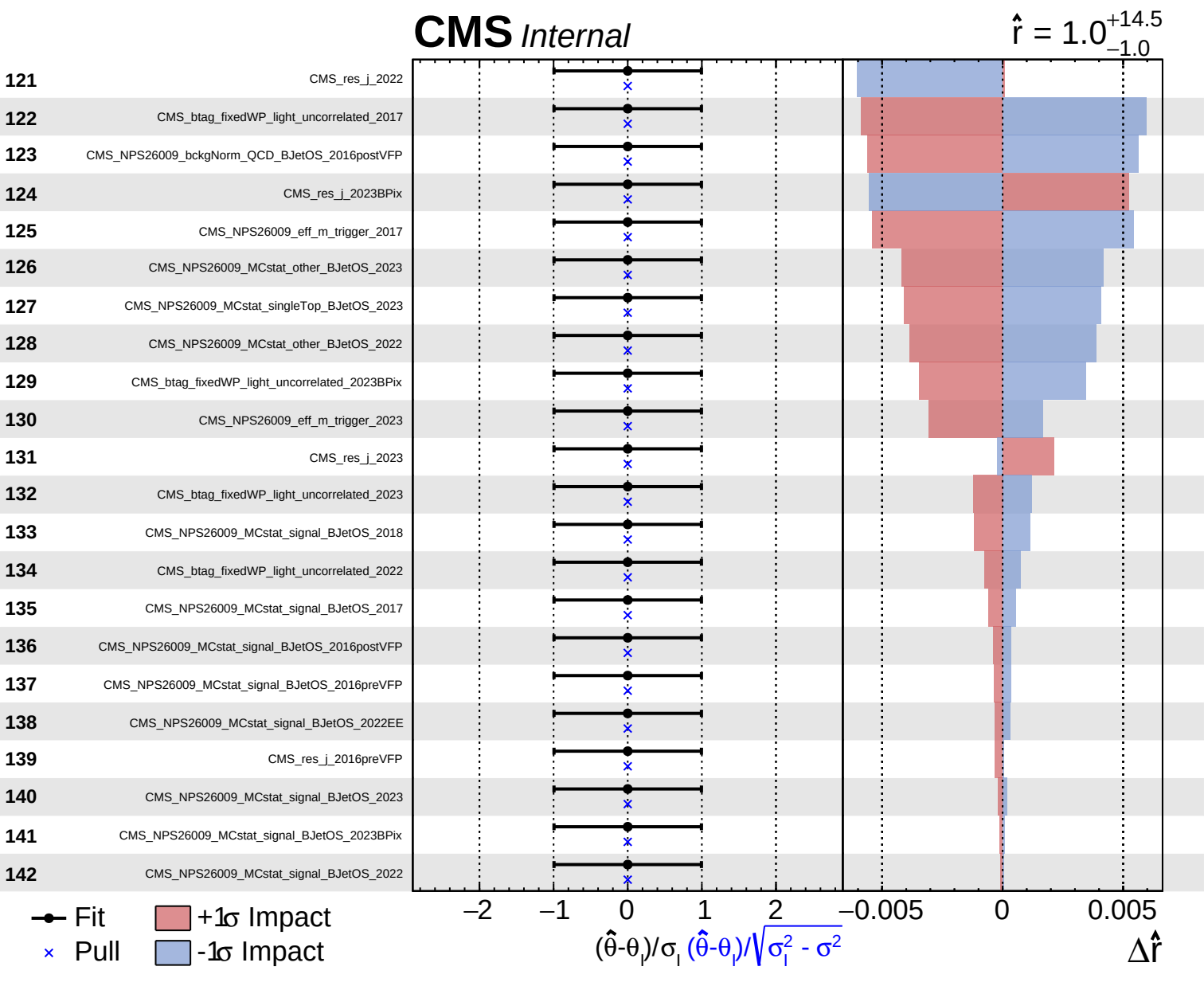

CMS Internal
$\hat{r} = 1.0^{+14.5}_{-1.0}$
1 QCDscale_ren_ttbar
2 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
3 CMS_NPS26009_QCD_MCTransferStat_Run2
4 CMS_NPS26009_NFStat_DY_BJetOS_2018
5 CMS_NPS26009_stat_QCD_BJetOS_2018
6 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
7 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
8 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
9 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
10 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
11 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
12 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
13 CMS_NPS26009_QCD_MCTransferStat_Run3
14 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
15 CMS_NPS26009_topmass_ttbar_BJetOS
16 QCDscale_fac_ttbar
17 pdf_alphas
18 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
19 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
20 CMS_NPS26009_stat_QCD_BJetOS_2022EE
21 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
22 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
23 CMS_NPS26009_stat_QCD_BJetOS_2022
24 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
25 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
26 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
27 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
28 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
29 CMS_NPS26009_pdf_NNPDF31
30 CMS_NPS26009_NFStat_DY_BJetOS_2023BPix
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−5 0 5
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+14.5}_{-1.0}$
31 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
32 CMS_NPS26009_NFStat_DY_BJetOS_2017
33 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
34 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
35 CMS_NPS26009_stat_QCD_BJetOS_2023
36 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
37 CMS_NPS26009_MCstat_other_BJetOS_2022EE
38 CMS_NPS26009_NFStat_DY_BJetOS_2022
39 lumi_13TeV_1516_l
40 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
41 lumi_13TeV_15161718_l
42 CMS_NPS26009_stat_QCD_BJetOS_2017
43 CMS_NPS26009_NFStat_DY_BJetOS_2023
44 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
45 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
46 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
47 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
48 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
49 CMS_NPS26009_QCDscale_ren_singleTop
50 lumi_1
51 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
52 CMS_btag_fixedWP_bc_uncorrelated_2018
53 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
54 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
55 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
56 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
57 CMS_scale_j_2018
58 CMS_pileup_13TeV
59 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
60 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.5
0
0.5
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+14.5}_{-1.0}$
61 CMS_btag_fixedWP_bc_uncorrelated_2022EE
62 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
63 CMS_pileup_13p6TeV
64 CMS_scale_j_2022EE
65 CMS_l1_prefiring_2016postVFP
66 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
67 CMS_l1_prefiring_2016preVFP
68 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
69 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
70 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
71 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
72 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
73 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
74 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
75 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
76 CMS_NPS26009_QCDscale_fac_singleTop
77 lumi_2
78 CMS_scale_j_2016postVFP
79 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
80 CMS_NPS26009_eff_m_id_Run2
81 CMS_scale_j_2016preVFP
82 CMS_scale_m_13TeV
83 CMS_NPS26009_MCstat_other_BJetOS_2018
84 CMS_btag_fixedWP_bc_uncorrelated_2022
85 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
86 CMS_l1_prefiring_2018
87 CMS_btag_fixedWP_light_uncorrelated_2018
88 CMS_scale_j_2023BPix
89 CMS_scale_j_2022
90 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.1
0
0.1
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+14.5}_{-1.0}$
91 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
92 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
93 CMS_btag_fixedWP_light_uncorrelated_2022EE
94 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
95 CMS_NPS26009_eff_m_trigger_2016preVFP
96 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
97 CMS_NPS26009_eff_m_id_Run3
98 CMS_NPS26009_eff_m_trigger_2018
99 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
100 CMS_btag_fixedWP_bc_uncorrelated_2023
101 lumi_13TeV_151617_l
102 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
103 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
104 CMS_l1_prefiring_2017
105 CMS_scale_m_13p6TeV
106 CMS_NPS26009_eff_m_trigger_2016postVFP
107 CMS_scale_j_2017
108 CMS_btag_fixedWP_bc_uncorrelated_2017
109 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
110 CMS_NPS26009_eff_m_trigger_2022EE
111 CMS_scale_j_2023
112 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
113 CMS_res_j_2016postVFP
114 CMS_res_j_2022EE
115 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
116 CMS_NPS26009_MCstat_other_BJetOS_2017
117 CMS_NPS26009_eff_m_trigger_2023BPix
118 CMS_NPS26009_eff_m_trigger_2022
119 CMS_res_j_2018
120 CMS_res_j_2017
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.04
−0.02
0
0.02
0.04
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+14.5}_{-1.0}$
121 CMS_res_j_2022
122 CMS_btag_fixedWP_light_uncorrelated_2017
123 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
124 CMS_res_j_2023BPix
125 CMS_NPS26009_eff_m_trigger_2017
126 CMS_NPS26009_MCstat_other_BJetOS_2023
127 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
128 CMS_NPS26009_MCstat_other_BJetOS_2022
129 CMS_btag_fixedWP_light_uncorrelated_2023BPix
130 CMS_NPS26009_eff_m_trigger_2023
131 CMS_res_j_2023
132 CMS_btag_fixedWP_light_uncorrelated_2023
133 CMS_NPS26009_MCstat_signal_BJetOS_2018
134 CMS_btag_fixedWP_light_uncorrelated_2022
135 CMS_NPS26009_MCstat_signal_BJetOS_2017
136 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
137 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
138 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
139 CMS_res_j_2016preVFP
140 CMS_NPS26009_MCstat_signal_BJetOS_2023
141 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
142 CMS_NPS26009_MCstat_signal_BJetOS_2022
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.005
0
0.005
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$